

### Question # 1:

- a. i)  $s \rightarrow r$       ii)  $\neg p \wedge (p \vee \neg q)$       iii)  $(r \wedge s) \leftrightarrow t$       iv)  $t \rightarrow ((r \wedge s) \vee \neg p)$

- b. Converse: "Flight is delayed whenever the assignment is not graded."  
Inverse: "The assignment is graded whenever flight is not delayed."  
Contrapositive: "Flight is not delayed whenever the assignment is graded."

c.

$$\begin{aligned} & \neg[r \vee (q \wedge (r \rightarrow p))] \\ & \equiv \neg r \wedge \neg(q \wedge (r \rightarrow p)), \\ & \equiv \neg r \wedge \neg(q \wedge (r \vee \neg p)), \\ & \equiv \neg r \wedge \neg(q \wedge (r \vee \neg p)), \\ & \equiv \neg r \wedge (\neg q \vee \neg(r \vee \neg p)), \\ & \equiv \neg r \wedge (\neg q \vee (\neg r \wedge p)), \\ & \equiv (\neg r \wedge \neg q) \vee (\neg r \wedge (\neg r \wedge p)), \\ & \equiv (\neg r \wedge \neg q) \vee ((\neg r \wedge \neg r) \wedge p), \\ & \equiv (\neg r \wedge \neg q) \vee (\neg r \wedge p), \\ & \equiv \neg r \wedge (\neg q \vee p), \\ & \equiv \neg r \wedge (p \vee \neg q), \end{aligned}$$

De Morgan's law  
Conditional rewritten as disjunction  
Double negation law  
De Morgan's law  
De Morgan's law, double negation  
Distributive law  
Associative law  
Idempotent law  
Distributive law  
Commutative law

$$\begin{aligned} d. & = (\neg s \rightarrow \neg h) \wedge (s \rightarrow w) \wedge (\neg w \wedge s) \\ & = (h \rightarrow s) \wedge (s \rightarrow w) \wedge (\neg w \wedge s) \\ & = (h \rightarrow w) \wedge (\neg w \wedge s) \\ & = (h \rightarrow w) \wedge (\neg w) \\ & = \neg h \end{aligned}$$

Law of Contrapositive  
Hypothetical Syllogism  
Simplification  
Modus Tollens  
Hence above arguments are valid.

e.

- i) Swimming at the New Jersey shore is allowed if and only if sharks have not been spotted near the shore.  
ii) If swimming at the New Jersey shore is allowed then sharks have not been spotted near the shore and conversely.  
iii) Swimming at the New Jersey shore is allowed is necessary and sufficient for sharks to not been spotted near the shore.

f. The statement is Contingency.

| a | b | c | $\neg a$ | $\neg b$ | $\neg c$ | $a \rightarrow b$ | $b \rightarrow c$ | $b \wedge \neg c$ | $((a \rightarrow b) \wedge (b \rightarrow c) \wedge (b \wedge \neg c))$ | $(a \rightarrow b) \wedge (b \rightarrow c) \wedge (b \wedge \neg c) \leftrightarrow \neg a$ |
|---|---|---|----------|----------|----------|-------------------|-------------------|-------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| T | T | T | F        | F        | F        | T                 | T                 | F                 | F                                                                       | T                                                                                            |
| T | T | F | F        | F        | T        | T                 | F                 | T                 | F                                                                       | T                                                                                            |
| T | F | T | F        | T        | F        | F                 | T                 | F                 | F                                                                       | T                                                                                            |
| T | F | F | F        | T        | T        | F                 | T                 | F                 | F                                                                       | T                                                                                            |
| F | T | T | T        | F        | F        | T                 | T                 | F                 | F                                                                       | F                                                                                            |
| F | T | F | T        | F        | T        | T                 | F                 | T                 | F                                                                       | F                                                                                            |
| F | F | T | T        | T        | F        | T                 | T                 | F                 | F                                                                       | F                                                                                            |
| F | F | F | T        | T        | T        | T                 | T                 | F                 | F                                                                       | F                                                                                            |

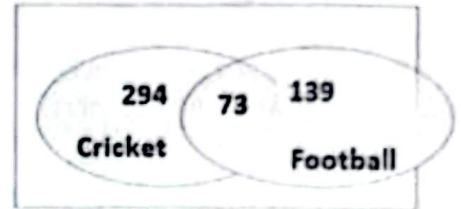
### Question # 2:

- a. i) There is a student in your university who has not connected with anyone on LinkedIn.  
 ii) Not every student in your university has connected with everyone on LinkedIn.

b. i)  $\forall x \exists y (S(x, y) \wedge C(y))$  ii)  $\forall x \forall y (S(x, y) \wedge A(y))$   $A(y) \rightarrow S(x, y)$

c. i) False ii)  $\neg \forall p \exists q \neg T(p, q) = \exists p \exists q p + 2q \neq pq$  True

d.  $|C \cup F| = |C| + |F| - |C \cap F| = 367 + 212 - 73 = 506$ .  
 Hence, 506 students play at least one game.



e.  $A \times B = \{(a, 1), (a, 2), (a, 3), (d, 1), (d, 2), (d, 3), (g, 1), (g, 2), (g, 3)\}$   
 $|A \times B| = 9 \rightarrow |P(A \times B)| = 2^9 = 512$

f.

|                       |                                                        |                                     |
|-----------------------|--------------------------------------------------------|-------------------------------------|
| $\overline{A \cap B}$ | $= \{x   x \notin A \cap B\}$                          | by defn. of complement              |
|                       | $= \{x   \neg(x \in (A \cap B))\}$                     | by defn. of does not belong symbol  |
|                       | $= \{x   \neg(x \in A \wedge x \in B)\}$               | by defn. of intersection            |
|                       | $= \{x   \neg(x \in A) \vee \neg(x \in B)\}$           | by 1st De Morgan law for Prop Logic |
|                       | $= \{x   x \notin A \vee x \notin B\}$                 | by defn. of not belong symbol       |
|                       | $= \{x   x \in \overline{A} \vee x \in \overline{B}\}$ | by defn. of complement              |
|                       | $= \{x   x \in \overline{A} \cup \overline{B}\}$       | by defn. of union                   |
|                       | $= \overline{A} \cup \overline{B}$                     | by meaning of notation              |